Joints

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joints

- **defined as places of union between two or more bones.**
- **Classified according to their structural features into fibrous, cartilaginous, and synovial types**.

Fibrous joints:

- bones are joined by fibrous tissue

-joints have no joint cavities or fibrous capsule.

- no movements allowed (minimal degree of movement may allowed in certain occasions)

- fibrous joints are presented by three types

1. Sutures(immobile)

- bones are connected by fibrous connective tissue.

- found between the flat bones of the skull.



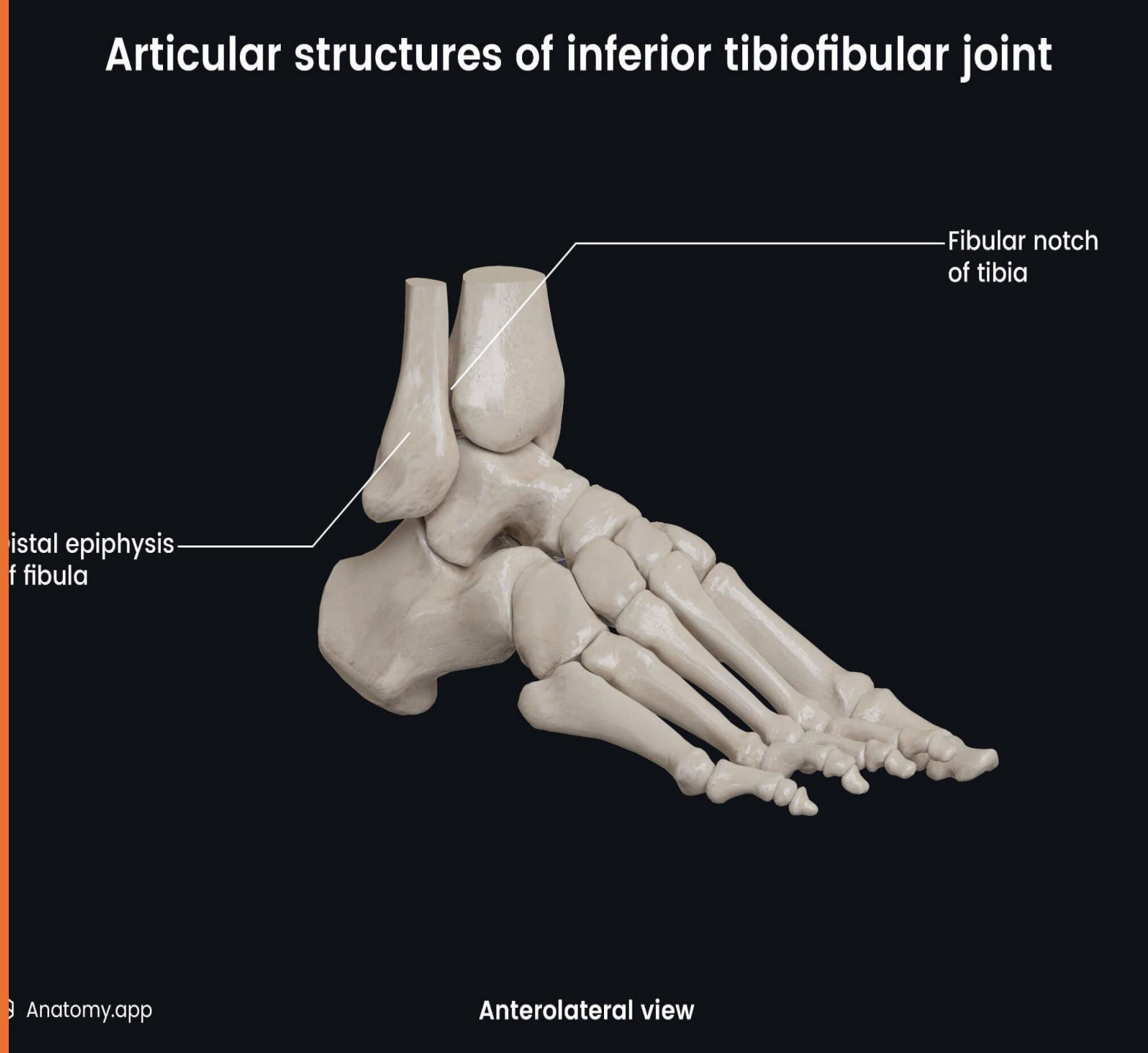


Fibrous joints:

2. Syndesmoses (slightly mobile)

- bones are connected by fibrous connective tissue described as an interosseus ligament.

- Occurs on the inferior tibiofibular joint, interosseus sacroiliac ligament, tympano-stapedial joint, interosseus membranes, ligamentum flavum ,supraspinous and interspinous ligaments of vertebrae and coracoclavicular ligament

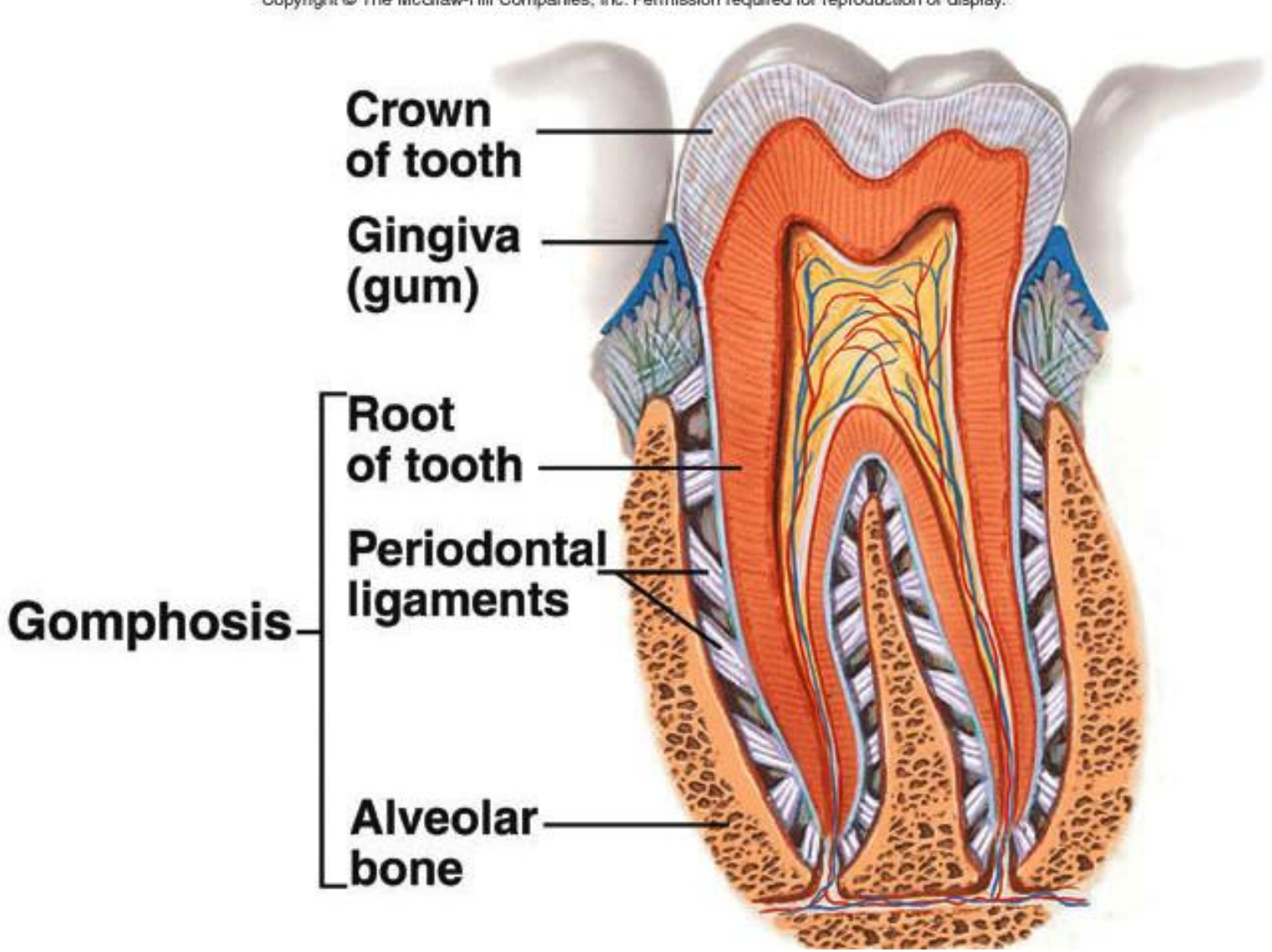


Fibrous joints:

3- gomphosis:

gomphosis is a fibrous mobile peg-and-socket joint.

The roots of the teeth (the pegs) fit into their sockets in the mandible and maxilla and are the only examples of this type of joint.



Cartilaginous joints

- bones are united by cartilage.

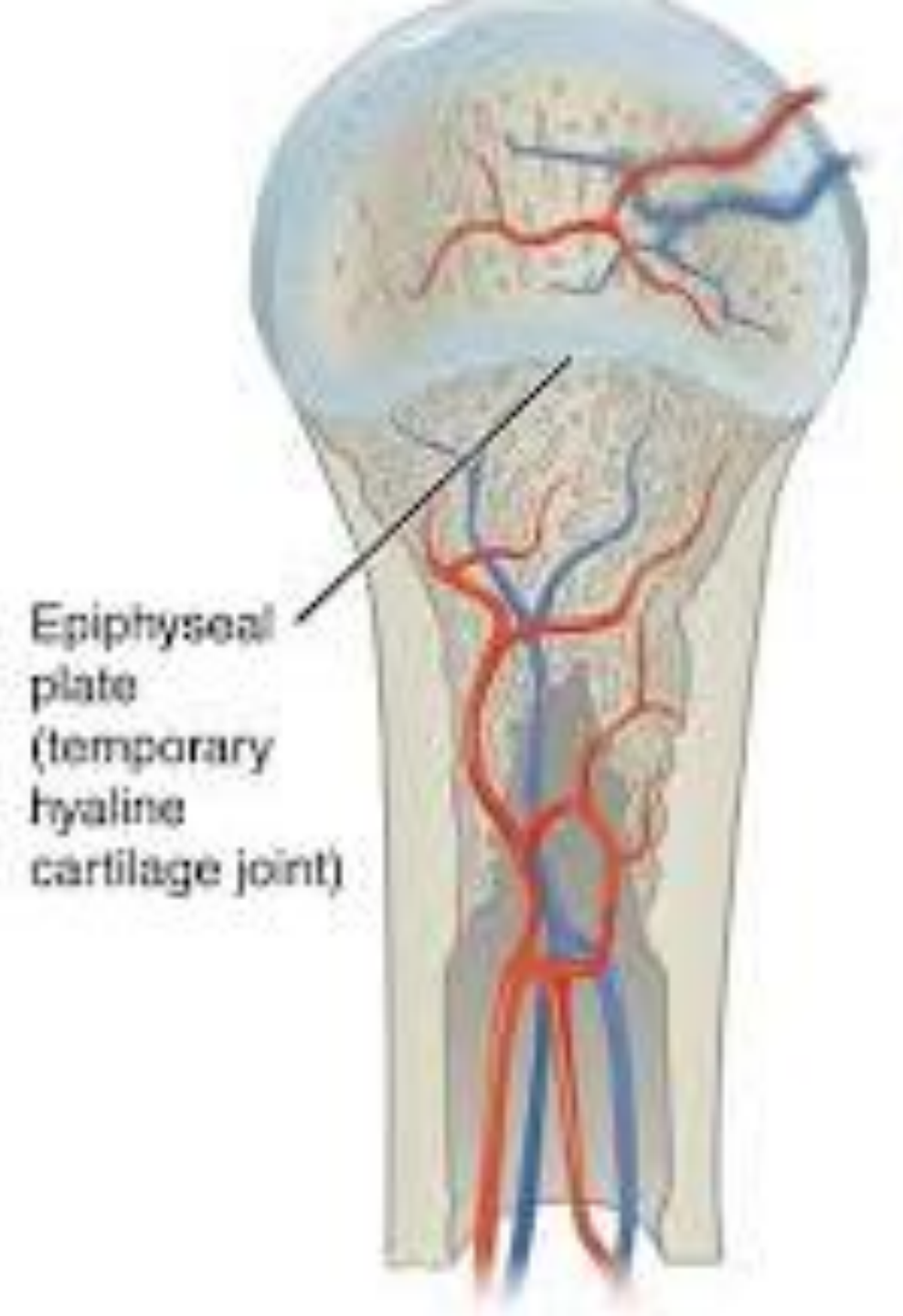
- have no joint cavity or fibrous capsule.

- according to type of cartilage, it can be classified into:

1. Primary Cartilaginous Joints (Synchondroses)

- bones are united by hyaline cartilage and permit little to no movement but allow for growth in length during childhood and adolescence.

- Include epiphyseal cartilage plates of long bones (the union between the epiphysis and the diaphysis of a growing bone) ,spheno-occipital and manubrio-sternal synchondroses, and triradiate acetabular cartilage of developing hip bone.



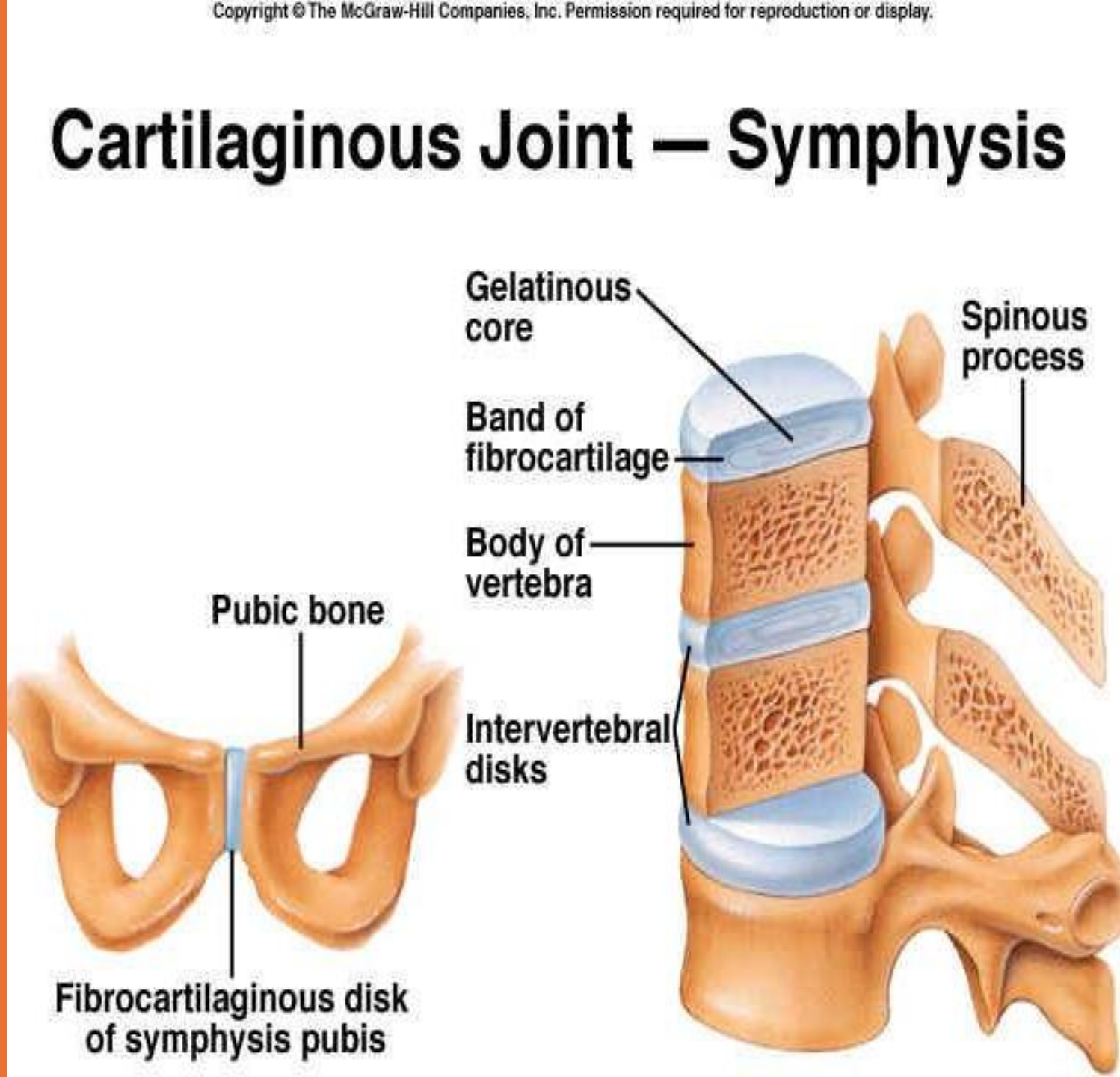
Cartilaginous joints:

2. Secondary Cartilaginous Joints (Symphyses)

- bones are joined by fibrocartilage and are slightly movable joints.

- Include the pubic symphysis , the intervertebral symphysis(discs), manubriosternal and xiphisternal symphysis.

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Synovial (Diarthrodial) Joints

- - Permit a great degree of free movement and are classified according to the shape of the articulation and/or the type of movement.
- - Are characterized by four main features: joint cavity, articular (hyaline) cartilage, synovial membrane (which produces synovial fluid), and articular capsule.

characteristics

- The synovial joints have the following characteristics:

a. articular surfaces: are covered by hyaline cartilage called articular cartilage and are separated from each other (except the clavicle, which is covered at sternoclavicular joint by white fibrocartilage)

b. fibrous capsule or articular capsule: the joint is surrounded by fibrous capsule which is attached to margins of articular surfaces and perforated by the vessels and nerves going to the joint. in certain joints, the fibrous capsule may contain openings for passage of tendons or synovial bursae, also it becomes thickened at its parts subjected to excessive strain to form what is called capsular ligaments

c. joint cavity: a "potential" joint cavity is formed

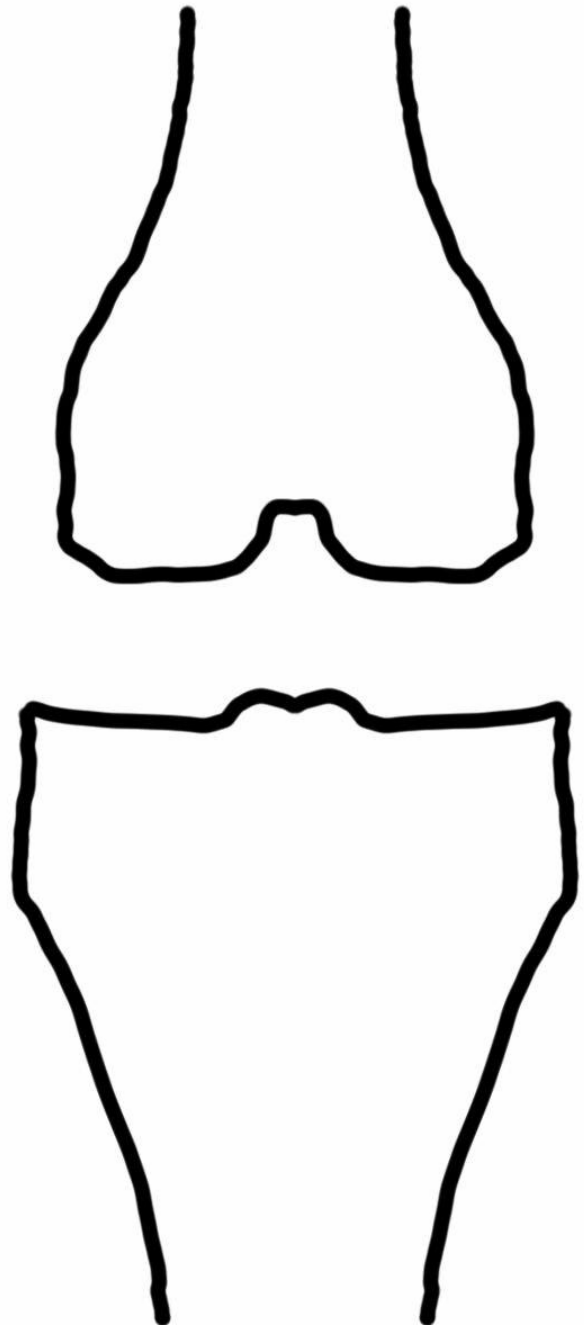
d. synovial membrane: lines the inner surface of the capsule and cover the non-articular parts of bones within the joint cavity, but never cover the articular cartilage.it secrets synovial fluid which lubricates, minimizes friction of the articular surfaces and nourish the articular cartilage which is avascular tissue by diffusion

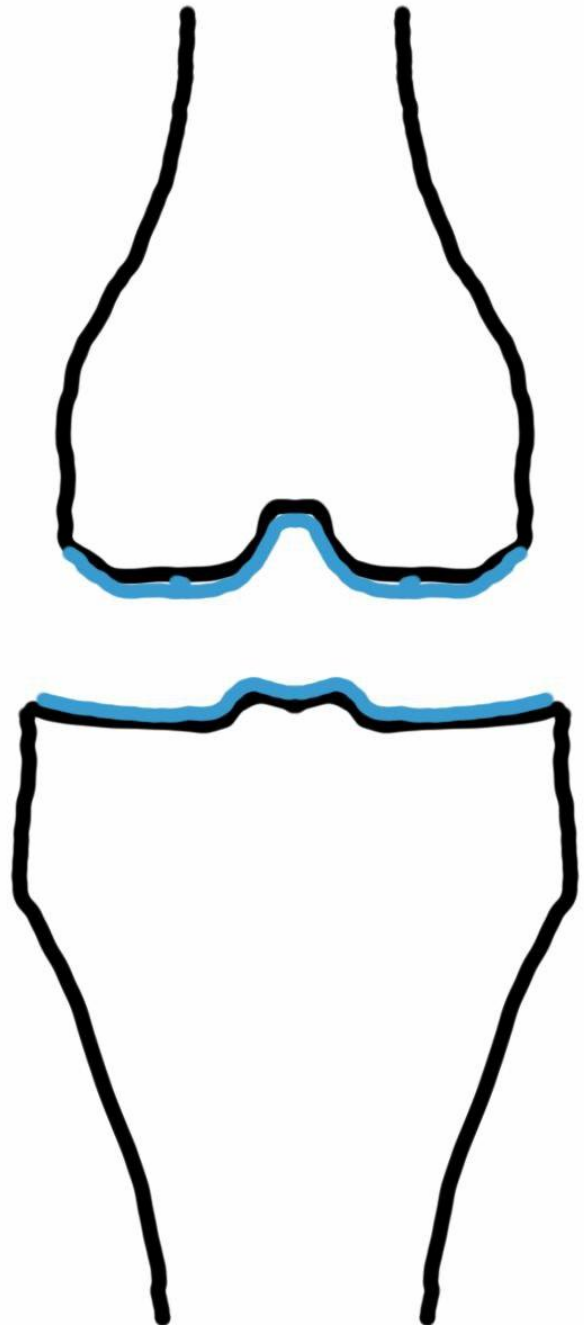
e. extracapsular ligaments: serve the going stability by holding the articular bones together, enforcing the capsule, limiting excessive movements and preventing undesired movements

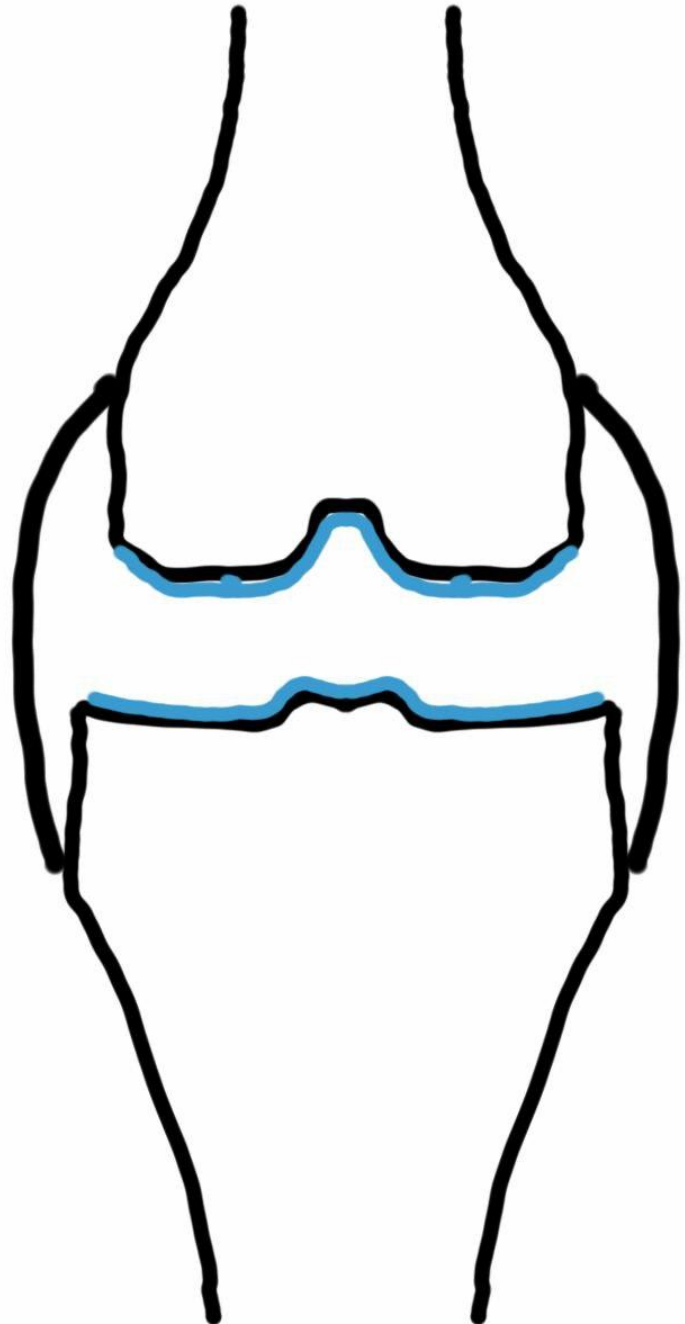
f. intraarticular or intracapsular structures: rather than the synovial membrane and synovial fluid which are intracapsular structures, some joints described as complex joints contain other intracapsular structures, which are:

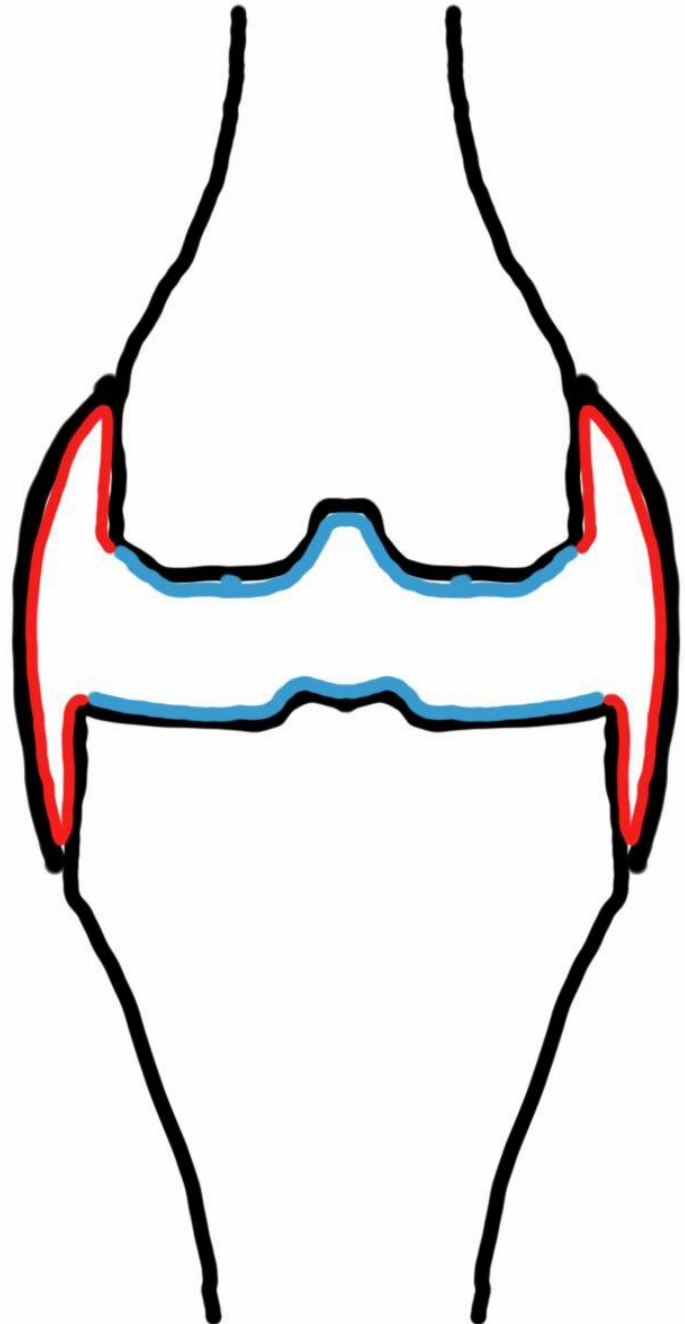
- articular disc: present in joints where gliding is associated with angular movements to minimize attrition of articular surfaces. It is present in; temporomandibular, sternoclavicular, acromioclavicular and wrist joints
- menisci; in the knee joint
- ligament; cruciate ligaments in the knee joint
- tendons; tendon of long head of biceps in shoulder and tendon of popliteus in knee joints.

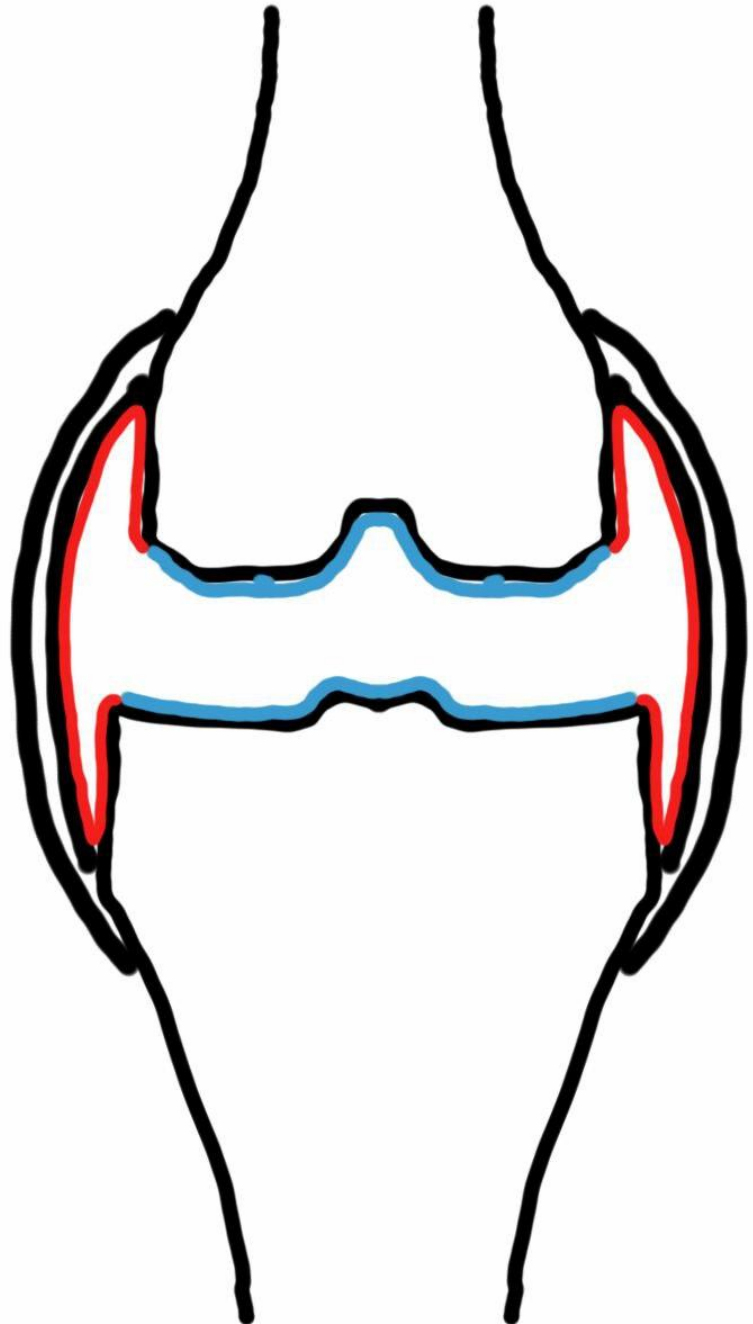
- Synovial joints have variable degree of mobility

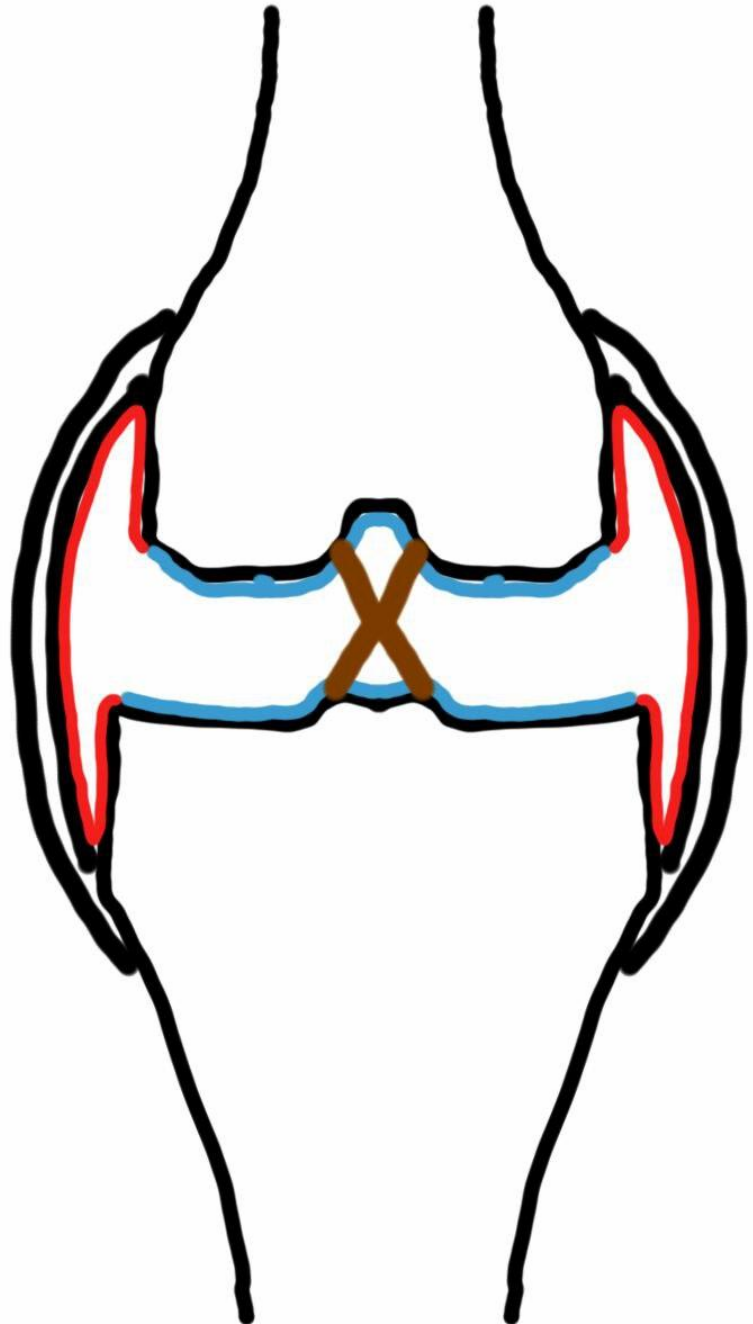


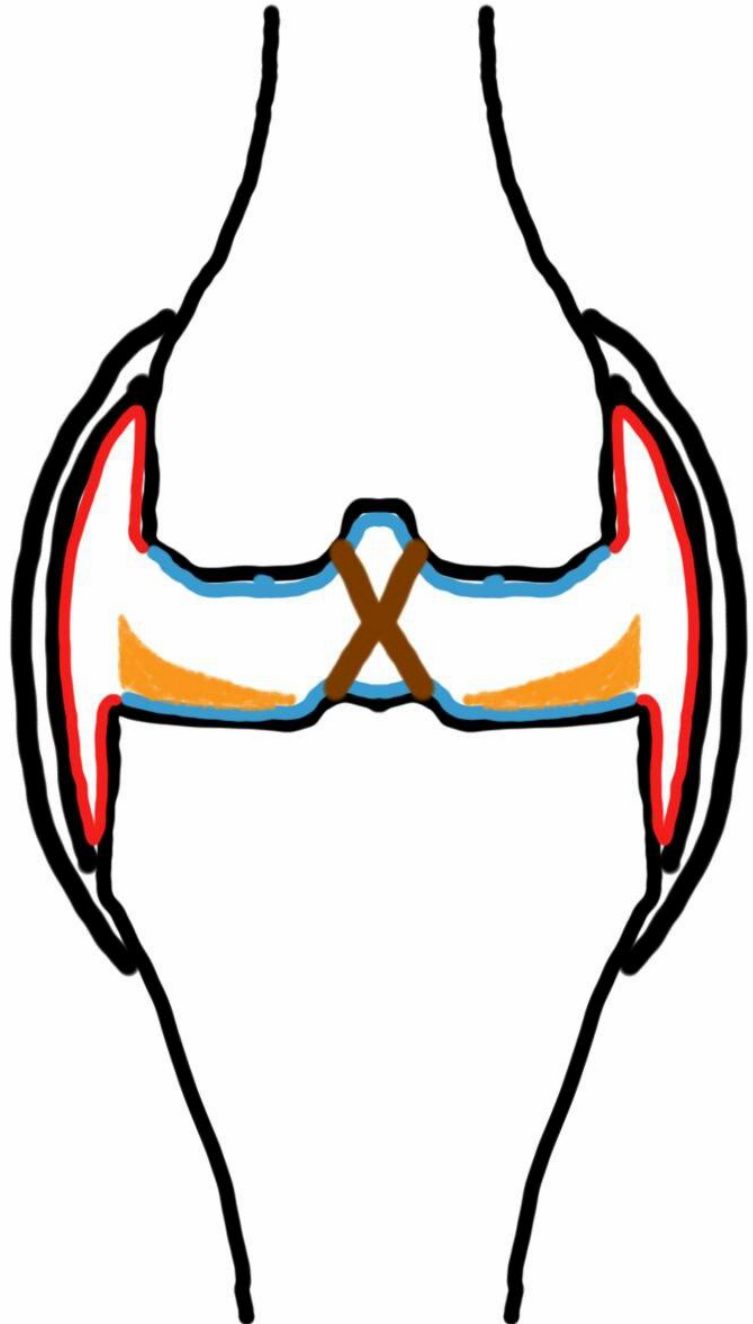












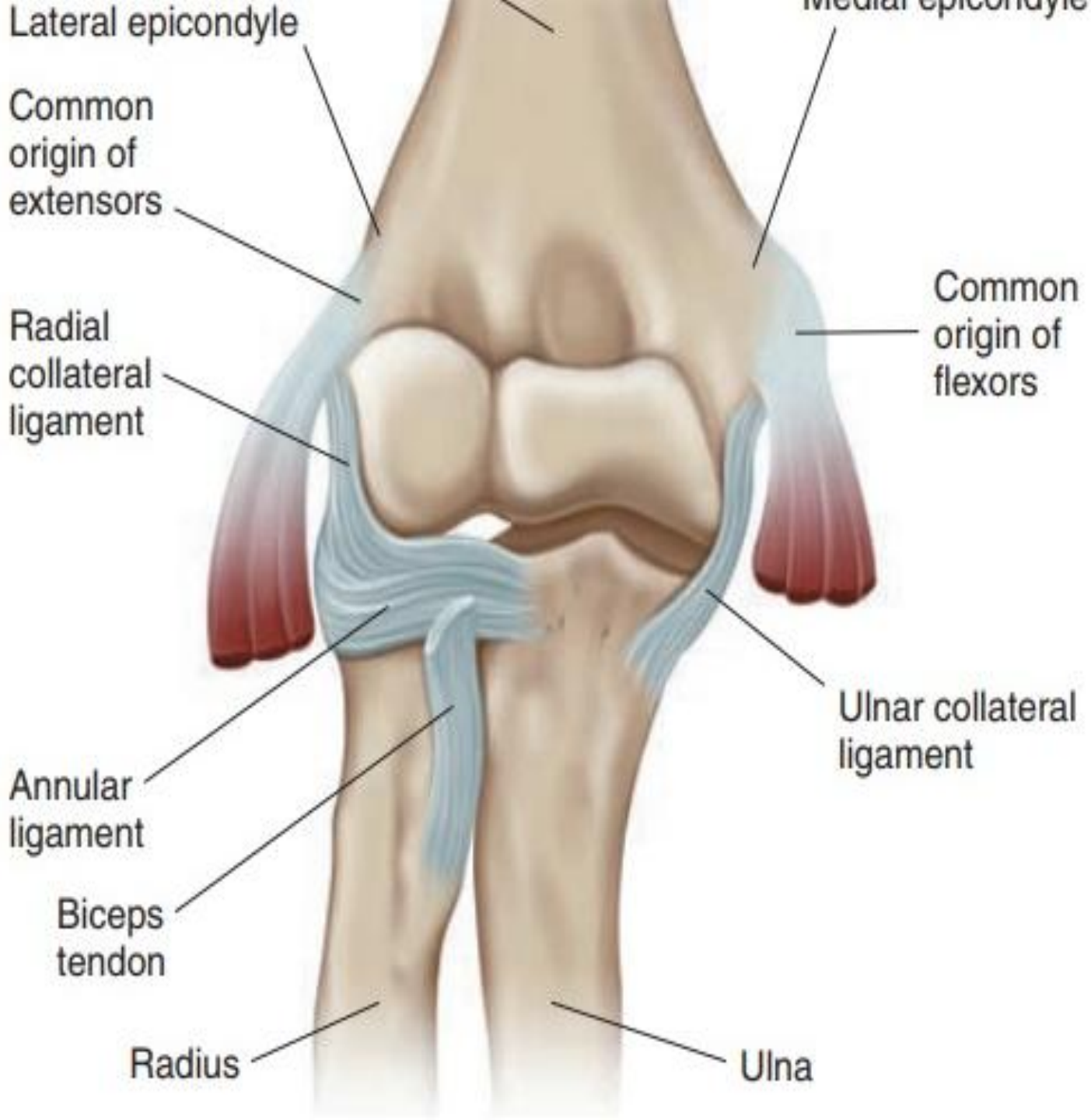
classification

According to shape of articular surfaces and according to number of axis of movements, synovial joints are classified into;

1. **Uniaxial synovial joints: movements occur around one axis and presented in two types:**

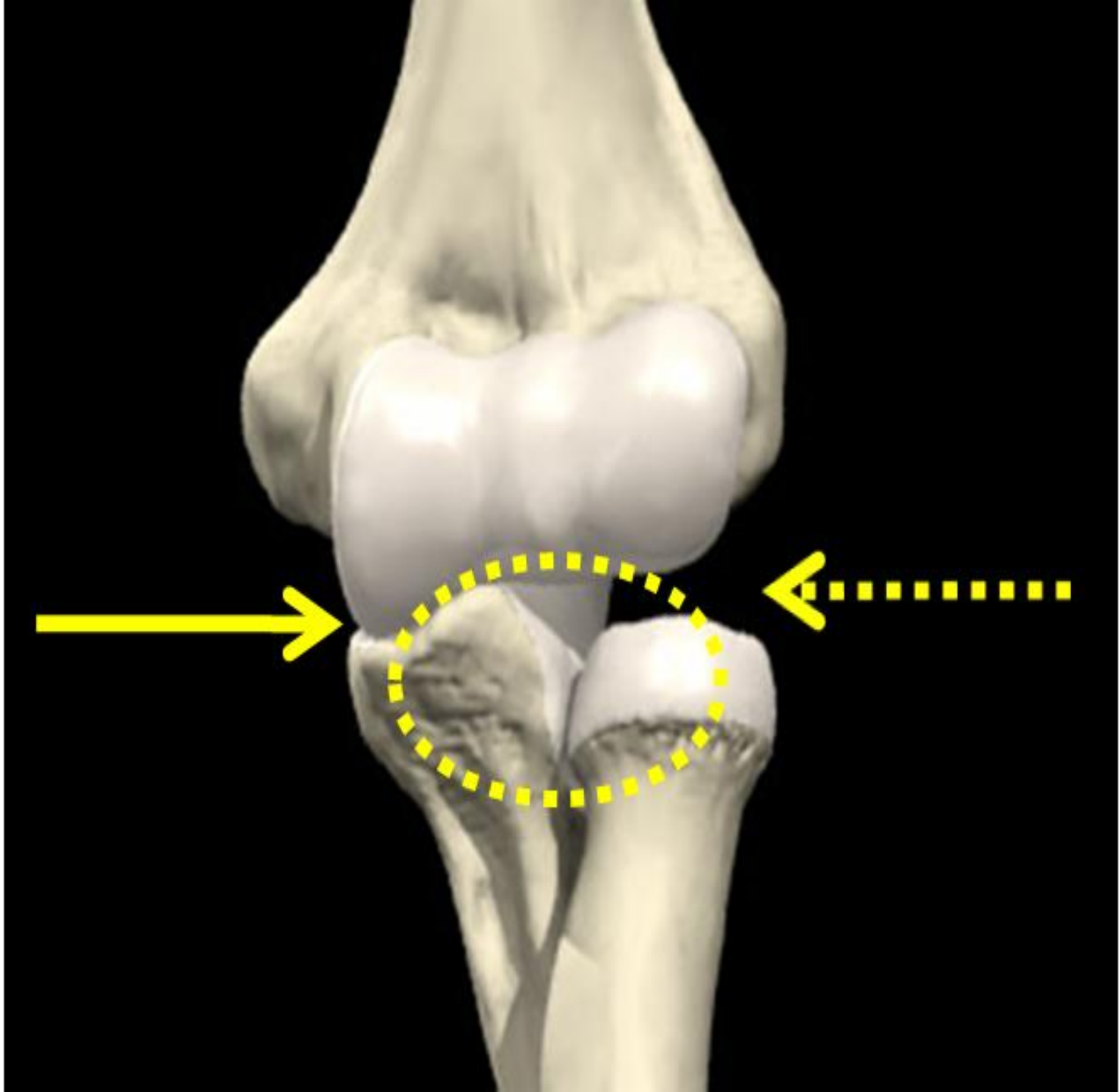
a. **Hinge (Ginglymus): movements occur around transvers axis as in elbow, ankle and interphalangeal joints**

b. **Pivot (Trochoid): formed of central bony pivot surrounded by ossiferous ring and allows only rotation.movements occur around longitudinal axis as in superior radioulnar and median atlantoaxial joints**



Pivot synovial

Ex.: superior radioulnar joint



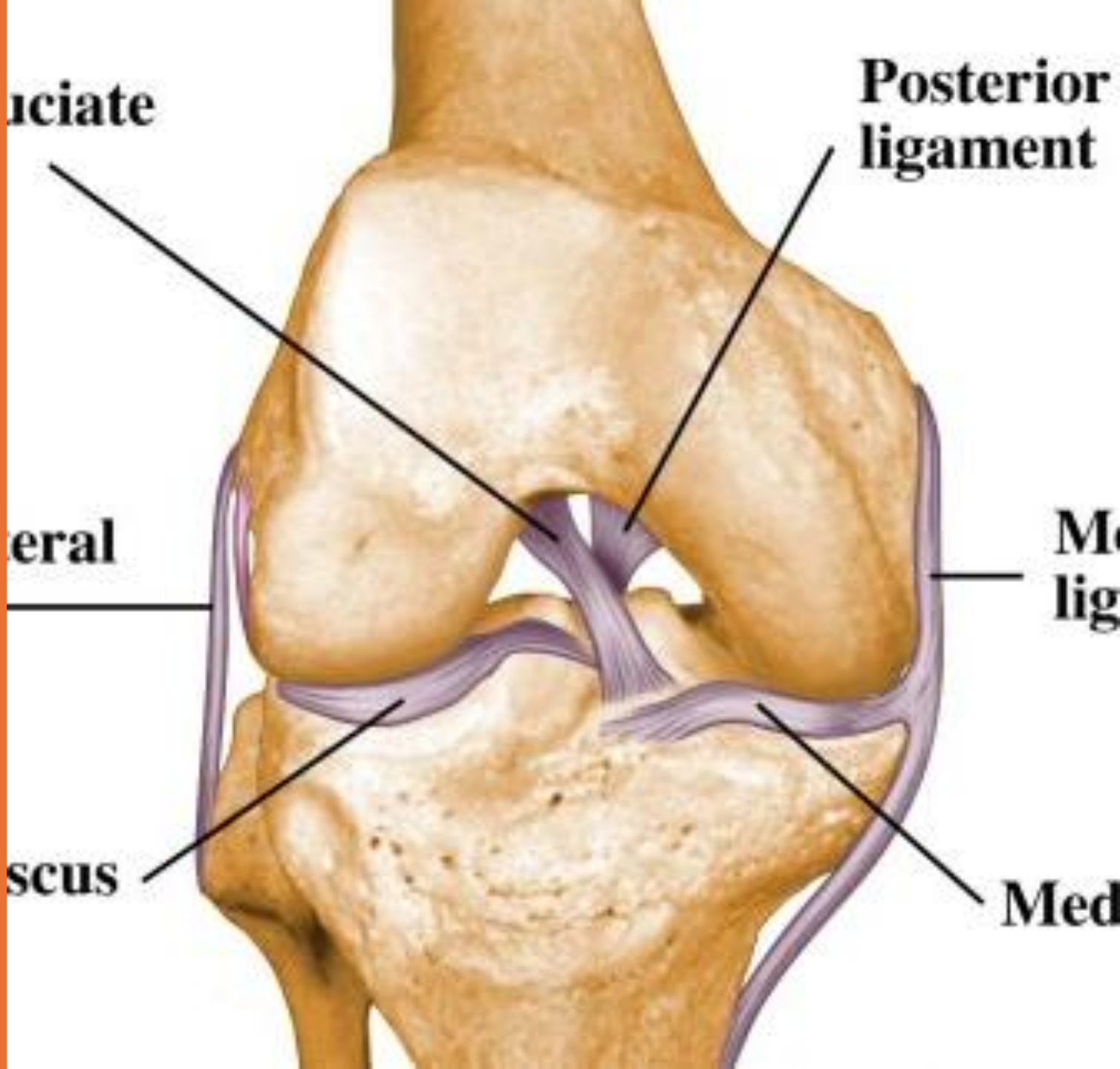
Classification:

2. Biaxial synovial joints: movements occur around two axes and presented in three types:

a. Bicondylar: The joint consists of two convex condyles articulating with two concave surfaces as in knee joint (common capsule) and temporomandibular joint (two separate capsules)

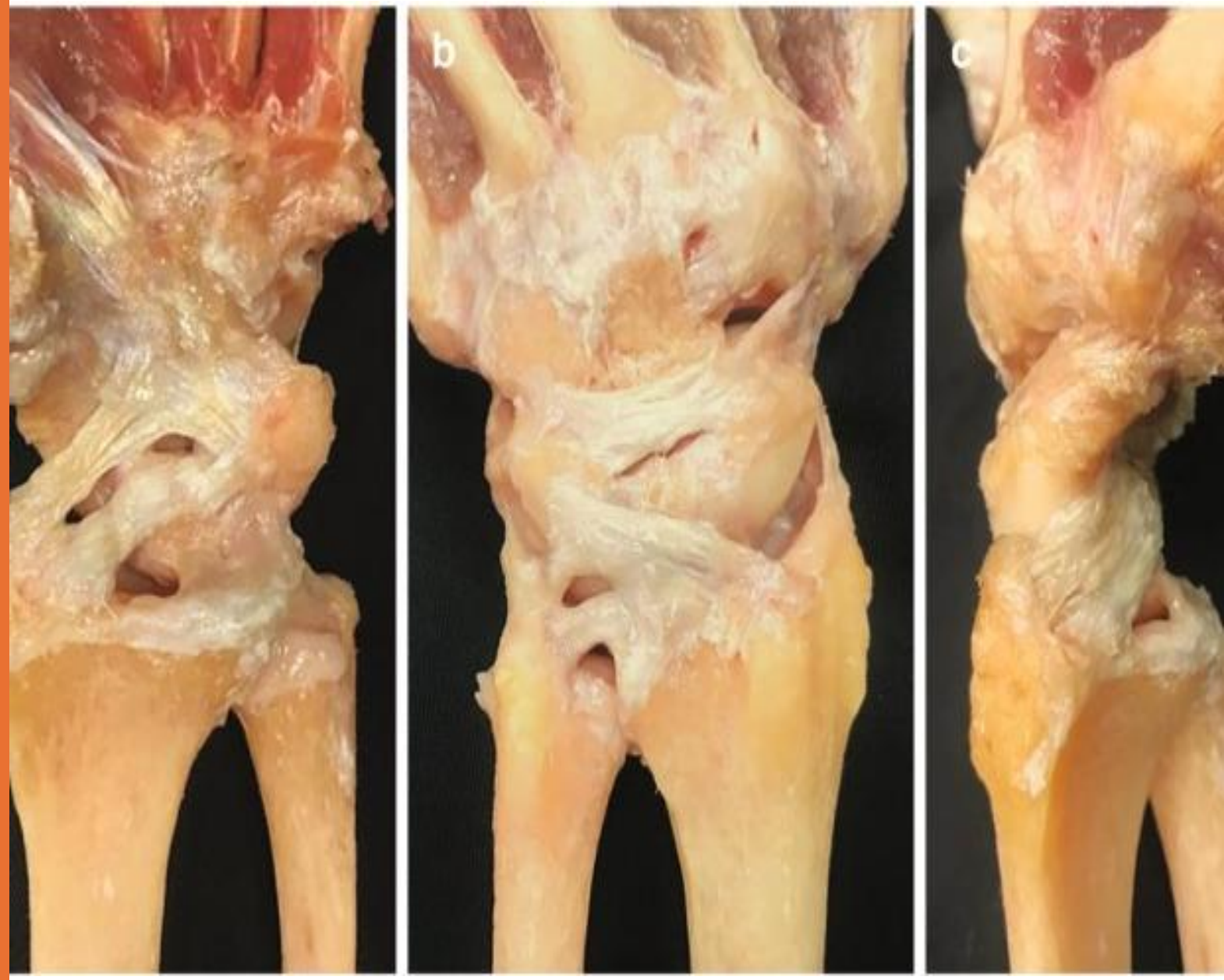
b. Ellipsoid(condyloid): one convex oval surface articulates with one concave oval surface as in wrist, metacarpophalangeal and metatarsophalangeal joints

c. Saddle (Sellar): the articular surfaces are reciprocally concavo-convex as in carpometacarpal joint of the thumb



Ellipsoid:

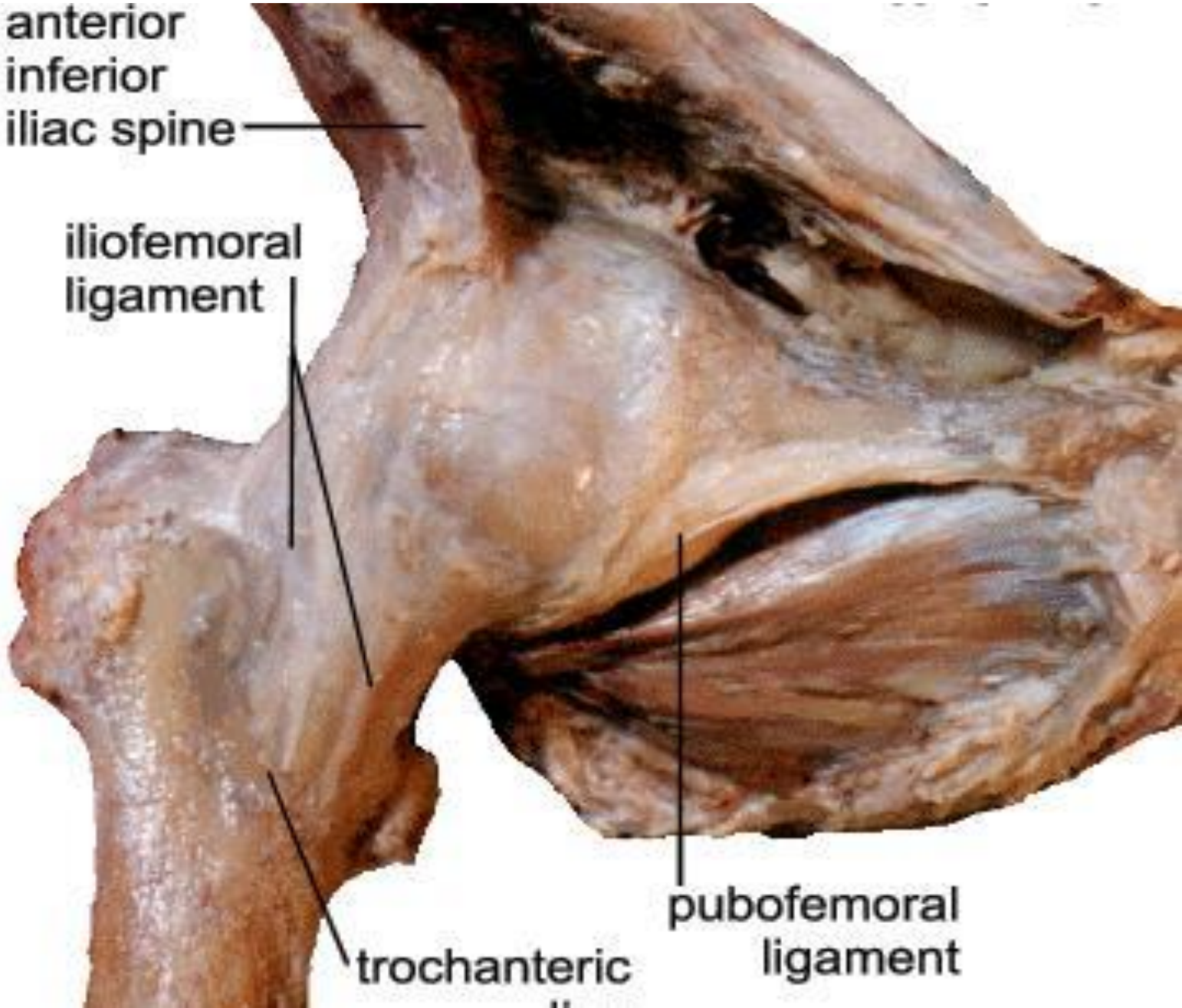
Ex.: wrist joint



Classification:

3- Polyaxial (called spheroidal or ball and socket or Cotyloid)): globular head is received into a cup-shaped cavity as in shoulder and hip joints.

4- Plane (Gliding) joints: articular surfaces are flat and nearly in apposition, as in facet (joints between articular processes of vertebrae), intertarsal, intercarpal, superior tibiofibular and acromioclavicular joint



Nerve supply of joints

- Are innervated as follows: The nerve supplying a joint also supplies the muscles that move the joint

and the skin covering the insertion of such muscles (Hilton’s law).